Structures

Group 8

March 18, 2026

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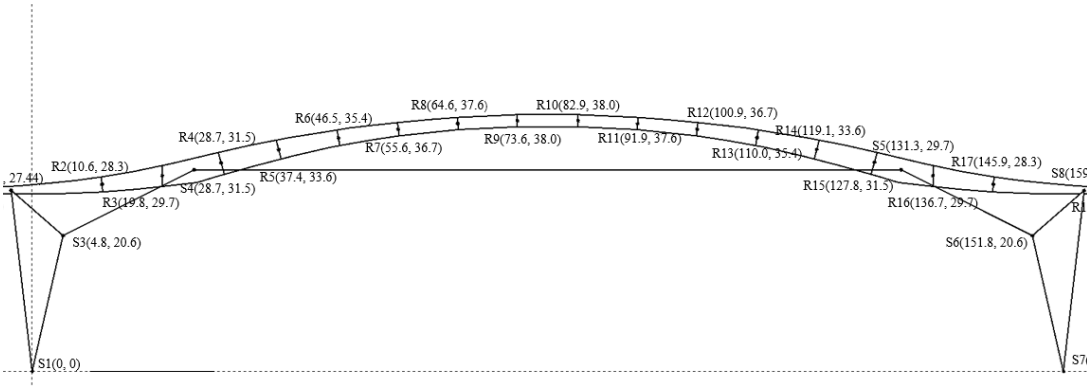
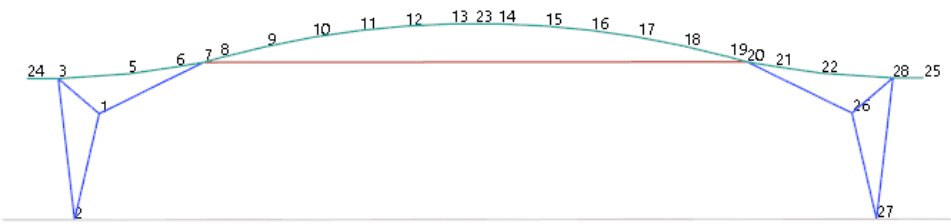
1 Deliverable 1: Geometry

The roof geometry was reconstructed from the supplementary drawing using a one-to-one SketchUp model and then transferred into GSA. As shown in Figure 1, points labelled R define the roof profile and represent the mid-points of the roof section, while points labelled S define the support system and lower restraint points. The reconstructed model was checked against the original drawing on both sides of the roof and the overall geometric deviation was approximately 0.5%, which is acceptable for global FE modelling.

The second diagram shows the FE idealisation adopted for analysis. In Figure 1(b), green denotes beam elements used for the roof profile, red denotes the high tie modelled as a tie element, and blue denotes the support-tree members modelled as bar elements. In GSA, beam elements model axial, bending and torsional effects, bar elements provide axial stiffness only, and tie elements are axial-only elements with zero stiffness in compression. This makes beams suitable for the roof rafters, bars suitable for the tree members, and a tie suitable for the prestressed high tie. The real high tie consists of twin locked-coil strands, but in the 2D global model these were idealised as one single tie element with the same total cross-sectional area. Both end supports were modelled as fixed supports, while the pin-connected members were treated consistently on both sides because the connected elements were assumed not to transfer bending moment. This idealisation captures the main load path and preserves the key dimensions required for later loading, validation and mesh sensitivity studies.



(a) Reconstructed nodal geometry



(b) Idealised FE member types

Figure 1: Geometry adopted for the global model.

2 Deliverable 2: Material, Section, and Loading Data

2.1 Material and Section Properties and Loading Data

The global FE model was defined using linear-elastic steel. In accordance with the brief, all members were assigned an elastic modulus of 200 GPa. A Poisson's ratio of 0.30 and density of 7850 kg m^{-3} were adopted so that self-weight could be included automatically in GSA. S355 steel was used for the primary roof members, which is consistent with the Heathrow T5 project information. The twin high-tie strands were modelled as a single prestressed tie element with the same total cross-sectional area and an initial prestress of 450 MPa.

Only vertical symmetric loading was considered. The characteristic permanent roof load was taken as the sum of the aluminium standing seam roof (0.10 kPa), structural deck (0.15 kPa), insulation/acoustic layers (0.40 kPa) and the service allowance (0.15 kPa), giving $G_k = 0.80$ kPa. The live load imposed on the roof was taken as $Q_k = 0.60$ kPa. The governing vertical ultimate limit state (ULS) combination used in the model was

$$1.35G_k + 1.5Q_k \tag{1}$$

Here, G_k and Q_k were converted to line loads using the projected tributary width carried by one transverse frame. The supplementary roof information states that the main rafters and associated tie-rods are spaced at 18 m centres, so a tributary width of 18 m was adopted for the 2D frame idealisation. On this basis, the superimposed permanent roof load is 14.4 kN m^{-1} (0.80×18) and the imposed live load is 10.8 kN m^{-1} (0.60×18), with member self-weight still included separately through the material density. This projected-area treatment is consistent with common roof-loading practice and with the basis commonly used for roof snow loading in Eurocode 1 [1].

The strap members were modelled as bar elements. Therefore, only their axial stiffness was represented in the global model, using an equivalent steel area of $A = 2 \times 500 \times 40 = 40\,000$ mm^2. The spacing between the two plates was neglected in this idealisation because the bar elements do not carry bending.

Table 1: Section data adopted for the global model

Member	Section	Modelling note
Arch rafter	Welded box, 800 mm wide, 50 mm wall thickness	Depth varies from about 1.8 m to 3.8 m along the roof profile
Tusk rafter	Welded I-section, 50 mm plate thickness	Overall depth varies from about 0.9 m to 3.8 m
Legs and arms	Circular Hollow Section (CHS) 914 mm × 40 mm	Taken directly from the supplementary sketch
Wings	CHS 711 mm × 50 mm	Taken directly from the supplementary sketch
Internal/external ties	CHS 457 mm × 40 mm	Thickness not stated in the source, so 40 mm was assumed
High tie	Twin locked-coil strands, 2 × 115 mm	Modelled as one equivalent tie element with the same total cross-sectional area; prestressed to 450 MPa
Straps	Equivalent rod area 40 000 mm^2	Based on two 500 mm × 40 mm plates; bending stiffness neglected

2.2 Assumptions

The assumptions made were the use of a linear-elastic S355 steel model, uniform wall thicknesses where indicated, a projected tributary width of 18 m for each transverse frame, automatic inclusion of self-weight through the material density, axial-only idealisation of the support-tree members, and a prestressed tie-element idealisation for the high tie.

3 Deliverable 3: Initial Model Validation

3.1 Assumptions

One important modelling assumption is that the real roof has been reduced from a 3D structural system to a 2D frame idealisation. This means only the in-plane behaviour of one transverse frame is represented directly, while out-of-plane bending, torsion and any load-sharing between adjacent frames are neglected. As a result, the model cannot capture twisting of the curved rafters, weak-axis bending caused by geometric eccentricity, or other local 3D effects associated with the actual connection arrangement. This simplification is acceptable for an initial global study because the loading considered here is vertical and symmetric, so the dominant response is expected to remain in-plane, but it should still be recognised as a source of modelling error when interpreting the validation results.

3.2 Model Validation

3.3 Model Verification

Because a converged answer to an incorrect model is still wrong, the following basic verification checks were successfully performed on the model to ensure that the solution process was correctly implemented:

- **Error Warnings:** The GSA solver completed the static analysis with no warnings regarding matrix singularities or instabilities.
- **Overall Equilibrium Check:** The sum of the vertical reaction forces (R_z) at the pinned supports was confirmed to equal the total applied vertical gravity loads ($\sum F = R \approx 0$).
- **Symmetry:** Because the structural geometry and the critical ULS load combination are perfectly symmetric, the resulting internal forces and deflected shape were verified to be entirely symmetric.
- **Qualitative Check:** The deflected shape aligns with physical expectations; the vertical loads cause the arch crown to displace downwards, putting the main ties into pure axial tension.

3.4 Model Adequacy

Taken together, the modelling assumptions, validation considerations and verification checks indicate that the global frame model is adequate for an initial study of the roof under symmetric vertical loading. However, it should still be understood as a simplified global model rather than a full representation of the three-dimensional structural behaviour.

4 Deliverable 4: Mesh Refinement Sensitivity Study

4.1 Introduction to Discretisation Error

Finite element analysis provides an approximate solution, so the discretisation adopted for the curved arch rafter must be checked before the results are used for interpretation. In this model, the arch is represented by straight beam elements between nodes; if too few elements are used, the idealised geometry no longer matches the true curvature and the predicted stiffness and internal force distribution can change significantly. To isolate this effect, refinement was applied only to the arch rafter, while the tusk rafter, support tree and high tie were kept unchanged. This ensures that the observed variation is caused by the arch mesh itself rather than by changing support conditions or joint modelling elsewhere in the frame.

4.2 Convergence Study Methodology

Three mesh densities were evaluated under the governing ULS vertical load combination ($1.35G_k+1.5Q_k+1.0P_s$). Mesh 1 is the coarsest faceted representation of the arch, Mesh 2 retains more intermediate points, and Mesh 3 preserves the full coordinate set taken from the supplementary geometry and is used as the reference model. To judge convergence, one displacement quantity and two force quantities were tracked: arch central deflection, high-tie axial force and arch central bending moment. These parameters give a useful picture of both overall stiffness and the redistribution of internal action as the curved geometry is refined.

Table 2: Influence of arch mesh density on key global responses

Mesh	Arch elements	DoF	Arch central Deflection (mm)	Tie Axial force (kN)	Arch central Bending moment (kNm)	% deflection change vs Mesh 3
Mesh 1	14	53	241.2	9725	12760	37.6%
Mesh 2	16	59	175.7	9309	960.2	0.2%
Mesh 3	22	77	175.3	9320	198.2	0.0%

A solution is considered converged when further refinement causes negligible change in the target quantity. The following strict convergence criterion was applied between iterations:

$$\frac{|\delta_n - \delta_{n-1}|}{\delta_n} < 2\% \tag{2}$$

Table 2 shows that the coarse arch discretisation significantly distorts the response. Using Mesh 3 as the baseline, Mesh 1 overestimates the arch central deflection by 37.6%, the high-tie force by 4.3% and the arch central bending moment by 6337.9%. Mesh 2 brings the deflection error down to only 0.2% and the tie-force difference to 0.1%, but the central bending moment is still 384.5% higher than in Mesh 3. This indicates that the global load path is already close to convergence by Mesh 2, whereas the bending response at the centre of the arch remains highly sensitive to how the curvature is represented. The very large percentage differences in bending moment are amplified by the small reference value in Mesh 3 (198.2 kN m), but they still show that a coarse geometric idealisation is not acceptable if bending results are to be used with confidence.

4.3 Conclusion

Based on these results, Mesh 3 is adopted for the remainder of the report. Although Mesh 2 is already adequate for deflection and tie force, only Mesh 3 captures the arch curvature accurately enough to stabilise the central bending moment, which is an important response quantity for interpreting the behaviour of the roof frame.

5 Deliverable 5: Influence of Prestressing Assumptions

5.1 Aim and Comparison Basis

The coursework brief asks for a comparison between the simplified assumption that prestress is applied after full erection and the actual Heathrow T5 construction sequence described in the supplementary paper. The paper states that the centre arch was assembled and prestressed at low level, the rafters were lifted clear of the temporary supports during stressing, the cable force was adjusted after a short period to account for creep, and the roof was then lifted and subjected to staged load transfer. The completed structure therefore reached its final position with locked-in forces already present.

Full staged-construction modelling was not carried out here. Instead, two simplified bounding models were compared using the same geometry, mesh density and vertical loading: one with prestress neglected and one with the prescribed 450 MPa prestress included. This gives a clear indication of how sensitive the final global response is to activation of the high tie.

5.2 Quantitative Comparison

Table 3: Simplified sensitivity of global response to high-tie prestress

Response quantity	**Prestress neglected**	**Prestress included**	**Change with prestress**
Maximum vertical displacement (mm)	490	175.3	−64.2%
Centre rafter bending moment (kNm)	9811	198.2	−98.0%
Tusk-to-arch connection bending moment (kNm)	16490	5837	−64.6%
High tie force (kN)	4296	9320	+116.9%
Horizontal reaction at supports (kN)	4211	3480	−17.4%

5.3 Discussion of Results

Using the updated values in Table 3, the influence of prestress is much stronger than in the earlier model. Including prestress reduces the maximum vertical displacement from 490 mm to 175.3 mm, a reduction of 64.2%. The centre-rafter bending moment falls from 9811 kNm to 198.2 kNm, which is a reduction of 98.0%, while the tusk-to-arch connection bending moment reduces by 64.6%. At the same time, the high-tie force rises from 4296 kN to 9320 kN, an increase of 116.9%, and the horizontal support reaction reduces by 17.4%. The structural behaviour therefore changes from a relatively bending-dominated frame response to a much more efficient arch-and-tie action once the high tie is activated.

5.4 Implications for Design and Tolerance

These updated results show that the prestress assumption is critical to both design interpretation and construction tolerance assessment. If prestress is neglected, the model greatly overestimates vertical deflection and rafter bending, but significantly underestimates the tensile force that must be carried by the high tie, its anchorage system and the connected support nodes. In other words, the controlling design actions move away from flexural demand in the rafters and towards force transfer in the tie system once prestress is included.

This conclusion is consistent with the supplementary construction sequence, where paired cable stressing, preset geometry, monitoring during stressing, staged load transfer and a final adjustment after creep were all used to control the final geometry and force state. The real roof is therefore expected to behave much closer to the prestressed model than to the unprestressed one. For the remainder of this report, the prestressed model is adopted as the minimum realistic representation of the completed roof, while recognising that a full construction-stage model would still be required if residual locked-in stress or erection tolerances were themselves critical outputs.

6 Deliverable 6: Boundary Conditions for More Detailed Analysis

For a more detailed FE study, the rafter splice between the tusk rafter and arch rafter should be extracted from the global frame model and analysed as a local submodel. The purpose of the local model would be to resolve the stress distribution in the splice plates, bolts or welds, and the adjacent box and I-section walls, which cannot be captured accurately by the beam idealisation used in the global model.

The local model should include a short length of member on each side of the splice so that the boundary conditions are applied away from the connection itself. This reduces artificial stress concentrations caused by restraining the model too close to the detail. A practical choice would be to include approximately one section depth of the arch rafter and one section depth of the tusk rafter beyond the splice on either side.

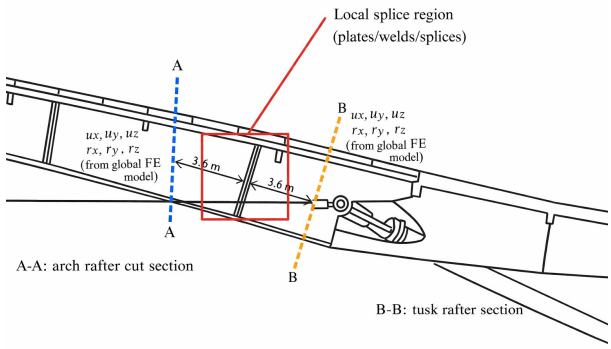


Figure 2: Recommended boundary conditions for the local splice model

The preferred boundary condition is to impose the displacement and rotation components extracted from the global beam model at the two cut faces of the local model. In practice, each cut face can be coupled to a reference node so that the sectional deformation remains compatible with the global model. The six kinematic quantities imposed at each end are the three translations (u_x, u_y, u_z) and the three rotations (r_x, r_y, r_z). This approach is preferable to simply fixing one end, because it reproduces the actual deformation state of the connection and avoids introducing artificial restraint moments.

The corresponding section forces should also be checked against the global model outputs. The main actions expected at the splice are axial force N, vertical shear V, bending moment M, and a smaller torsional component due to the curved roof geometry and eccentricity of the connection detail. If the software or modelling approach makes direct displacement transfer difficult, an acceptable alternative is to apply the equivalent set of sectional resultants at the two cut faces, provided that overall equilibrium is maintained.

To prevent rigid body motion in the local model, one reference node may be lightly stabilised after the global displacements have been assigned. This stabilisation should only remove numerical singularities and must not alter the transferred load path. Contact, bolt preload, or weld representation can then be added depending on the intended level of detail. For the purposes of this coursework, the key requirement is that the splice boundaries are driven by the global model response rather than by arbitrary fixed supports.

Section cuts are located approximately one local section depth away from the splice connection along the axis of the member. This avoids local stress disturbance. As the depth varies along the rafter, the local section depth is equal to the splice-region depth (3.6 m), cuts A–A and B–B are placed approximately 3.6 m from the splice on either side. At both cut faces, the translational and rotational displacements extracted from the global FE model are imposed.This ensures compatibility between the local splice model and the global structural response.

References

[1] British Standards Institution. BS EN 1991-1-3:2003+A1:2015 Eurocode 1: Actions on structures. General actions. Snow loads, 2015.